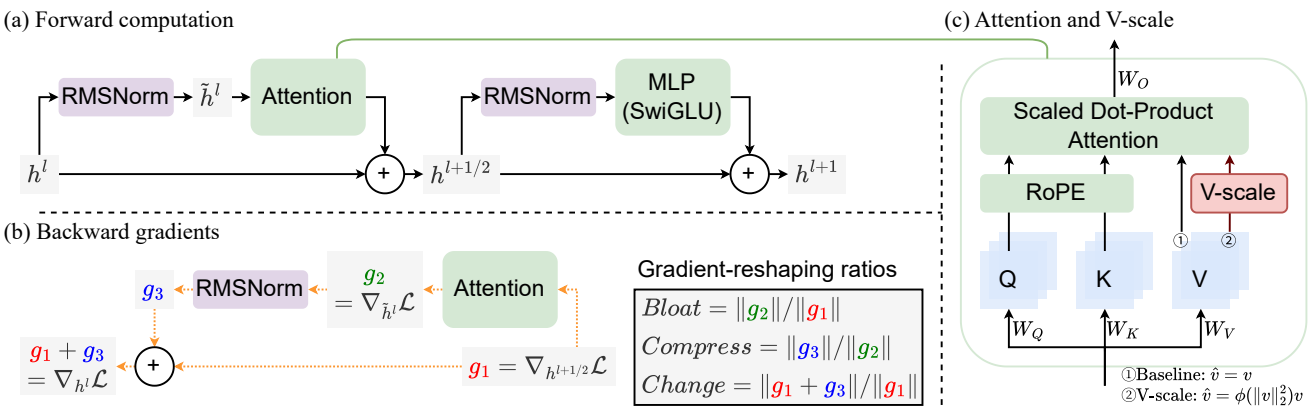

(a) Forward computation
RMSNorm
$\tilde{h}^l$
Attention
RMSNorm
MLP
(SwiGLU)
h^l
$h^{l+1/2}$
h^{l+1}
(b) Backward gradients
g_3
RMSNorm
g_2
$= \nabla_{\tilde{h}^l}\mathcal{L}$
Attention
$g_1 + g_3$
$= \nabla_{h^l}\mathcal{L}$
$g_1 = \nabla_{h^{l+1/2}}\mathcal{L}$
Gradient-reshaping ratios
$Bloat = \|g_2\|/\|g_1\|$
$Compress = \|g_3\|/\|g_2\|$
$Change = \|g_1 + g_3\|/\|g_1\|$
(c) Attention and V-scale
W_O
Scaled Dot-Product Attention
RoPE
V-scale
①
②
Q
K
V
W_Q
W_K
W_V
①Baseline: $\hat{v} = v$
②V-scale: $\hat{v} = \phi(\|v\|_2^2)v$